

Hyperscaler Overbuild, On-Premise SLM Substitution and the Anthropic Case Study

Final Report: an experimental research viewpoint

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1. What Is This Report About?

It is rare for a jockey to ask for the race to be slowed while his horse is out in front. So when the heads of the two leading AI labs call on the industry to pace itself [11], it is fair to ask why. The stated reason is safety, and it may be the whole story. This report examines another possibility that could sit alongside it: that the economics of the race have become harder than the headline growth figures suggest.

For the last year, the message from the AI industry was: spend more, generate more, use more, and you will be rewarded. When companies tried this, many found that costs rose faster than productivity gains. The Associated Press reported in July 2026 that “tokenmaxxing” was hitting its limits for exactly this reason, citing a Moody’s Ratings report that recommends a more disciplined approach, and noted that Microsoft CEO Satya Nadella had warned that customers are paying twice for AI: once in tokens and again by handing over their proprietary data [26]. Security risks, practicality, return on investment and productivity gains are now central questions in many company boardrooms [25][26]. This has slowed the growth of token spending at many enterprises, which are moving from “tokenmaxxing” to token budgets [25][27]. At the same time, the frontier labs are leasing data-center capacity built by others, from SpaceX and potentially from Meta, as builders look for buyers for capacity they have already paid for [3][7][8]. In our view, the data-center spending cycle is approaching its peak.

This report focuses on the usage trend at Anthropic, which Menlo Ventures ranks as the number one AI model provider by enterprise LLM API spend, with an estimated 40% share [16]. Readers should note that Menlo Ventures is itself an Anthropic investor [17].

For the past 24 months, global equity markets and hyperscaler balance sheets operated under the premise of unconstrained “tokenmaxxing”: the conviction that enterprise software would perpetually funnel compounding token volumes through very large foundation models hosted in centralized mega-clusters. Swerve Research Technologies presents network telemetry measured from a single user’s vantage point, together with published research, third-party usage data and public financial data, which in our view suggest that this premise is weakening. Enterprises are restructuring AI budgets toward efficiency, cheaper open-weight models and, increasingly, smaller task-specific models [22][25][27]. We believe this raises the risk of overcapacity among infrastructure builders and of monetization pressure on frontier model providers.

The Meta overbuild and compute dilemma

In late July 2026, Meta Platforms raised the low end of its 2026 capital expenditure guidance, to a range of \$130B–\$145B [1], nearly double its 2025 capital expenditure of about \$72B [55]. In Q2 2026, Meta’s free cash flow fell to \$784M from \$8.55B a year earlier, a decline of about 91% [1][2]. Meta has acknowledged that it might build more data centers than its AI products currently require [3], although Zuckerberg has also said Meta currently uses all of its capacity internally [6]. In June 2026, Anthropic approached Meta about leasing capacity; the talks, first reported by The New York Times, concern a deal worth up to \$10 billion over two years, with an early-exit option for either party [3][4]. The talks are early, and a source told CNN that the reported figures are speculative [5]. On Meta’s Q2 earnings call, Zuckerberg said Meta had received offers for its compute at a meaningful premium over what it had invested to build that capacity [1]. In September, after Anthropic’s CEO called for an industry-wide slowdown, Zuckerberg rejected a coordinated slowdown, arguing that competition and liability give each lab the incentive to act safely and that labs can pace themselves [13][14][15].

Hyperscaler cash flow and fixed-commitment exposure

Across the hyperscaler cohort, capital intensity has separated from free cash generation. Alphabet raised its 2026 capex guidance to \$195–\$205 billion and reported negative quarterly free cash flow (–\$5.9 billion in Q2 2026) for the first time since its IPO [56]. Microsoft guided to roughly \$175 billion of reported calendar-2026

capex; this is lower than its earlier guidance of about \$190 billion only because of a lease-accounting reclassification, and the company says its underlying investment plan is unchanged [57].

Frontier AI developers also carry large compute commitments. Anthropic agreed to pay SpaceX \$1.25 billion per month through May 2029, nearly \$45 billion in total, although either party can end the agreement with 90 days' notice [8], and Elon Musk has said SpaceX could pull back the capacity if it becomes tight [9]. Anthropic has also announced capacity agreements with Amazon (up to 5 GW), Google and Broadcom (5 GW, coming online from 2027) and Microsoft and NVIDIA (\$30B of Azure capacity) [10], and is in talks with Meta [3]. Its total computing infrastructure spend reportedly rose from \$3.4B in Q1 2026 to \$5.6B in Q2 [41]. Meanwhile, Anthropic's CEO has publicly called on the industry to slow the pace of capability improvement [11].

2. Empirical Grounding: Local SLM Parity (Saad-Falcon et al., Stanford University)

A key reason enterprises are reconsidering centralized LLM spending is documented in research by Saad-Falcon, Narayan and colleagues at Stanford University and Together AI [18]. The study evaluated more than 20 local language models with 20 billion or fewer active parameters, including Qwen3, Gemma3, gpt-oss and IBM Granite 4.0, on about one million real-world single-turn chat and reasoning queries, running on local hardware such as the Apple M4 Max and comparing them with frontier cloud models [18].

Table 1. Key findings from the Stanford local-inference research

Finding	Result	Source
Single-turn chat and reasoning queries answered accurately by local models	88.7%	[18]
Share of queries the best single local model can serve (win/tie vs frontier)	23.2% (2023) → 48.7% (2024) → 71.3% (2025)	[18][19]
Intelligence-per-watt improvement, 2023–2025	5.3× (3.1× from models, 1.7× from hardware)	[18][19]
Best-of-local routing vs best-of-cloud routing	Local ahead on 3 of 4 benchmark categories	[18]
Hybrid local–cloud routing with an 80%-accurate router vs cloud-only	64.3% less energy, 59.0% lower cost	[18]
Local vs data-center hardware running the same model (Qwen3-32B)	M4 Max ≈1.5× lower intelligence per watt than NVIDIA B200	[18]
Best on-device configuration vs Claude Opus 4.6 on personal-AI agent tasks (2026)	Qwen3.5-122B: 80.3% vs 83.5% (within 3.2 points)	[20]
Small 9B local model swapped in for Claude Opus 4.6 in existing agent stacks	Accuracy drops 25–39 points	[20]

Core findings

- **Most everyday queries do not need a frontier model.** OpenAI's own usage research, cited by the Stanford authors, shows that about 77% of ChatGPT requests are practical guidance, information seeking or writing, tasks that do not require frontier capabilities [18]. Local models already answer 88.7% of single-turn chat and reasoning queries accurately [18].
- **The gap is closing fast.** The share of queries the best local model can serve rose from 23.2% in 2023 to 71.3% in 2025 [18][19]. In 2026 research, the best on-device configuration came within 3.2 points of Claude Opus 4.6 on personal-AI agent tasks [20].

- **Cost and energy savings come from routing.** In the authors' simulation, a router that correctly sends 80% of queries to the right local or cloud model cut energy use by 64.3% and cost by 59.0% compared with cloud-only inference; perfect (oracle) routing would cut them by 80.4% and 73.8% [18]; see also [21]. Per watt, data-center hardware remains more efficient than local hardware running the same model [18]; the savings come from using smaller models, not from local chips being more efficient.
- **Limits.** The 2025 study excluded agentic tool use, long-horizon planning and long document processing, where local models lag frontier models by up to 45 percentage points [18]. Small local models still fall well short on agentic tasks [20]. The substitution threat is therefore greatest for simple, high-volume workloads.
- **Enterprise direction.** Gartner predicts that by 2027 organizations will use small, task-specific AI models at least three times more than general-purpose LLMs, citing faster responses and lower compute, operating and maintenance costs [22].

3. Production LLM Serving Traces and Campus Network Traffic

Two peer-reviewed measurement studies inform our methodology and our reading of serving economics.

ServeGen production serving trace (Xiang et al., Peking University and Alibaba Group; USENIX NSDI '26)

The authors characterized 3.54 billion requests across 12 models on Alibaba Cloud Model Studio over four months [23]. Relevant findings:

- **Bursty, uneven demand.** Short-term request arrivals for language models are often bursty (coefficient of variation greater than 1), with strong daily cycles in which load peaks in the afternoon and drops sharply in the early morning [23]. Reasoning-model workloads were less bursty [23]. Capacity sized for peaks therefore sits partly idle off-peak.
- **Heavy client concentration.** In one workload, the top 29 of 2,412 clients generated 90% of requests [23]. When demand is this concentrated, a change in behavior by a handful of large customers, such as adopting caching, token budgets or cheaper models, can move total load substantially.
- **Multimodal and reasoning overhead.** Half of image requests spent 75% of their time-to-first-token on download, normalization and encoding before the model began prefill [23]. Reasoning outputs contained on average about four times as many reasoning tokens as answer tokens [23].
- **Provisioning.** In one production case study, workloads generated naively from simple traces led to 50% under-provisioning compared with real workloads [23], meaning real traffic is harder to serve than simple benchmarks suggest.

UNSW campus network measurement (Lyu, Wang and Sivaraman, AINTEC '24)

This study was the first to track generative-AI usage through real-time analysis of network traffic, identifying flows by the TLS Server Name Indication (SNI) field on a full mirror of a university's campus traffic [24]. The authors explicitly note the method's value to financial investors needing a pulse on market trends [24]. Over five months (September 2023 to March 2024) across seven platforms, they found that the median session for every platform contained a single prompt-response pair, and that Claude's median user waiting time was about 4 seconds, with a median server response time of about 3 seconds [24]. ChatGPT dominated campus usage, with 54,448 prompts compared with 83 for Claude; GitHub Copilot usage tracked assessment deadlines; and ChatGPT usage rose at the start of the third semester [24].

4. Industry Re-Platforming: The Structural Migration to On-Premise SLMs

Across major enterprise verticals, architecture decisions are shifting toward smaller, cheaper and more controllable models. The pace varies widely by industry.

Banking and capital markets. Financial institutions operate under data-protection regimes such as GLBA, OCC and FINRA rules in the US and GDPR in Europe, which subject the sending of customer records and trade data to multi-tenant AI services to strict controls. Large banks are building domain-specific models; JP Morgan, for example, developed its own financial text analysis model [28]. Gartner expects task-specific models to be used at least three times more than general-purpose LLMs by 2027, particularly in domains such as financial services, healthcare and manufacturing [22][28]. Microsoft's CEO has warned that customers of frontier models are "paying twice", in tokens and in the proprietary data they hand over [26].

Automotive. Safety-critical driving functions require low, deterministic latency and must work without connectivity, so they run on in-vehicle compute platforms such as NVIDIA DRIVE Thor and Qualcomm Snapdragon Ride. In-cabin voice assistants, however, are moving the other way: Mercedes-Benz's MBUX Virtual Assistant uses Google Cloud's Automotive AI Agent, built on Gemini running on Vertex AI [29]. Automotive is therefore a mixed case: on-device for driving, cloud for conversation.

BPO, call centers and customer service. Operating under tight margins, customer-service providers are sensitive to per-token costs. Frontier output tokens on Anthropic's API are priced at \$15–\$25 per million (Claude Sonnet 5 at \$15 since its \$10 introductory price ended on 31 August 2026; Claude Opus 5 at \$25) [36][37], while leading Chinese open-weight models such as MiniMax M2.5 cost \$0.30 per million input tokens and \$1.10 per million output tokens, roughly 17 to 23 times cheaper per token than Claude Opus 4.6 (\$5/\$25) [30]. AI startup Lindy moved all of its Claude traffic to DeepSeek to control costs [25], and some enterprises now cap AI spending per employee [27]. On OpenRouter, Chinese models have taken more than 30% of weekly token use by US companies every week since February 2026, peaking at 46% [31].

5. Retail Saturation and OpenAI's Low-Cost Tier Commoditization

The assumption that consumer subscriptions could absorb excess capacity is being tested. OpenAI launched its low-cost ChatGPT Go plan in India at roughly \$4–\$5 per month in August 2025 and rolled it out globally in January 2026 at \$8 per month, alongside the \$20 Plus plan [34]. OpenAI has since introduced advertising on its Free and Go plans in the US (February 2026), made text chats uncapped for Free and Go users (rolling out from August 2026), and cut API prices, with GPT-5.6 Luna at \$0.20/\$1.20 per million input/output tokens from 30 July 2026 [34]. In our view, the combination of a lower-priced tier, ad-supported access and falling API prices indicates that consumer and developer AI is commoditizing. That leaves model developers more dependent on enterprise and API billing: roughly 80% of Anthropic's revenue is estimated to come from enterprise and API customers [35].

6. Empirical Study: Anthropic Case Study

A. Cloud partner and infrastructure architecture

Anthropic serves commercial and developer workloads through several channels:

1. **Amazon Web Services (AWS Bedrock).** A major commercial distribution channel. Anthropic has announced an agreement with Amazon for up to 5 GW of capacity, including nearly 1 GW of new capacity by the end of 2026 [10]. Anthropic's reported gross margins exclude revenue shared with distribution partners such as Amazon [43].
2. **Google Cloud Platform (Vertex AI).** Serves Claude to enterprise customers. Anthropic has announced a 5 GW agreement with Google and Broadcom coming online from 2027 [10].
3. **Anthropic first-party (Claude API, Claude.ai and Claude Code).** Anthropic's direct developer platform and subscription products.
4. **Leased and partner compute.** Anthropic agreed to pay SpaceX \$1.25B per month through May 2029 for Colossus capacity, terminable by either party on 90 days' notice [8]; is in talks to lease up to \$10B of capacity from Meta over two years [3]; and has announced \$30B of Azure capacity with Microsoft and NVIDIA and a \$50B US infrastructure investment with Fluidstack [10].

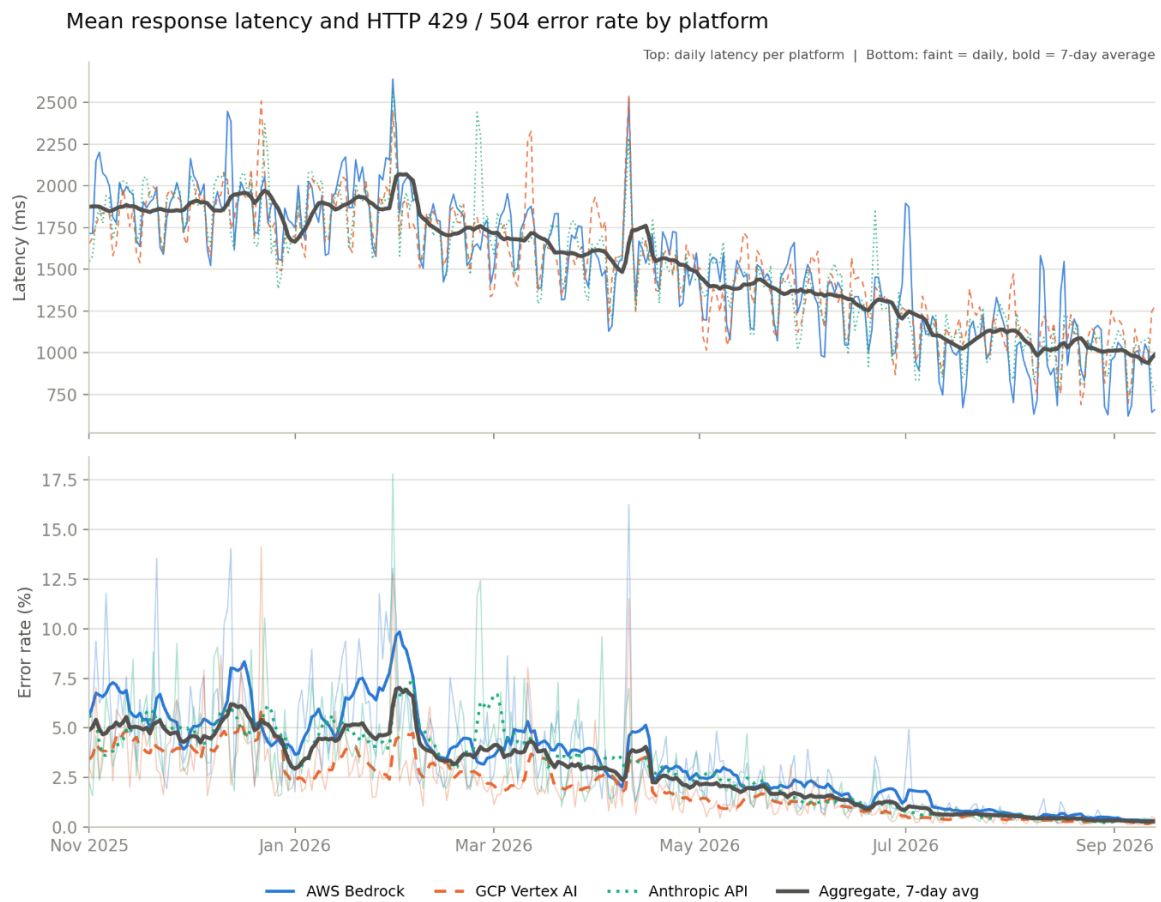
B. Observed telemetry: latency, errors and congestion

We measured response latency, time to first token and HTTP error rates from a single user's vantage point on requests sent to three platforms that serve Anthropic models (AWS Bedrock, Google Cloud Vertex AI and the Anthropic API), in order to analyse when it is best to run a larger workload on the servers that offer an Anthropic endpoint. We then used the measured latency to compute a congestion index; the only model involved is the M/D/1 queueing model used to convert measured latency into an estimated utilisation index (see Notes on Methodology). Because the data come from one user's vantage point, other users measuring from their own accounts, regions or networks may observe different values.

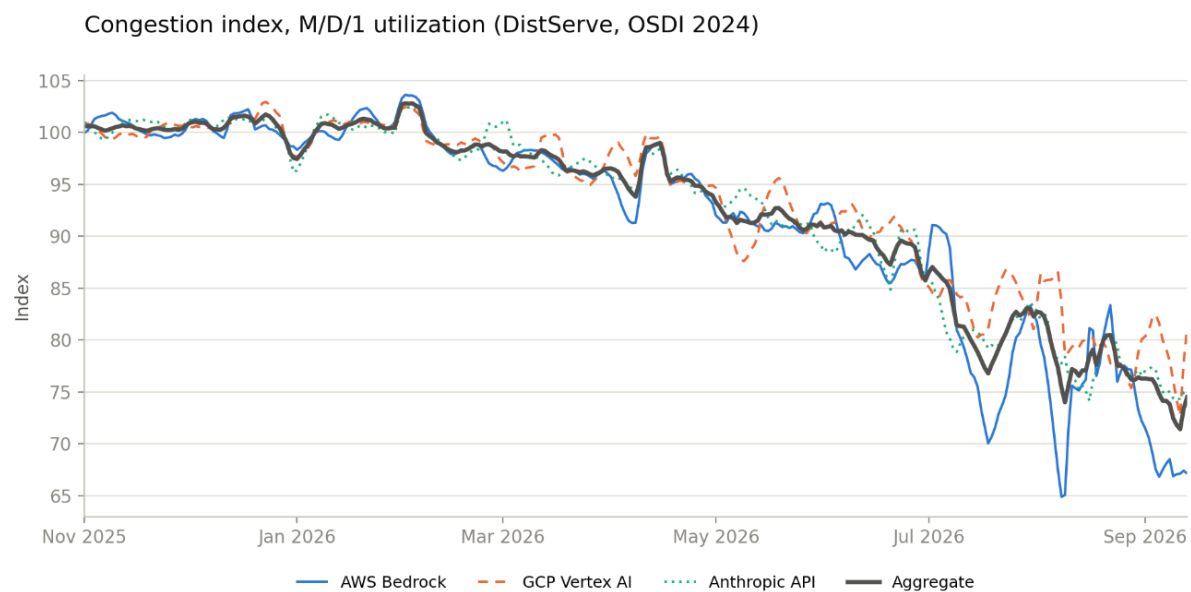
Figure 1. Summary of measured telemetry and derived congestion index, January vs September 2026

Metric	Jan 2026	Sep 2026	Change	What it indicates
AWS Bedrock congestion index (M/D/1, Jan-Feb 2026 = 100)	100.6	68.1	−32.3%	Lower estimated queueing on the observed Bedrock path
Google Cloud Vertex AI congestion index	100.5	79.0	−21.4%	Lower estimated queueing on the observed Vertex path
Anthropic API congestion index	100.3	75.6	−24.6%	Lower estimated queueing on the observed Anthropic API path
Aggregate congestion index	100.4	74.2	−26.1%	Requests served with less queueing delay
Aggregate estimated utilization (ρ)	0.83	0.61	−26.1%	Estimated share of capacity in use under the M/D/1 model, relative to capacity
Mean response latency	1,920 ms	980 ms	−49.0%	Faster responses for this observer; consistent with lower congestion, added capacity, or faster models
Error rate	5.1%	0.3%	−94.1%	Fewer errors for this observer; partly reflects the observer's own quota

Source: Swerve Research measurements from a single user's vantage point; congestion index and utilisation (ρ) derived with the M/D/1 queueing model of time to first token in DistServe [51]. Index: January–February 2026 = 100. Error rate partly reflects the observer's own quota and rate-limit tier.

Figure 2. Mean response latency and HTTP 429/504 error rate by platform, November 2025 to September 2026

Source: Swerve Research measurements from a single user's vantage point. Top panel: daily mean latency per platform, with the aggregate 7-day average. Bottom panel: faint lines are daily values; bold lines are 7-day averages. Other observers may record different absolute latency and error rates.

Figure 3. Congestion index based on M/D/1 utilisation, November 2025 to September 2026

$\rho = 2q / (1 + 2q)$, where $q = W / D - 1$, W is the measured 7-day average latency and D is the base service time (546 ms, fitted to the measured latency series in Figure 2 and scaled per platform). Index: January–February 2026 = 100. The queueing model follows DistServe (Zhong et al., OSDI 2024) [51]. The underlying latency data are measured from a single user's vantage point; only the index is model-derived.

C. What do these graphs mean?

Consider the requests made to the servers (the POST messages) as cars, and the servers themselves as highways. Congestion is the traffic caused when the number of cars increases and the highways are not wide enough. So when does congestion ease? There are three main reasons why congestion eases and the cars (requests) can travel faster:

- 1) **Better traffic management**, the equivalent of server optimisation.
- 2) **Bigger highways**, the equivalent of capacity addition.
- 3) **Fewer cars**, the equivalent of a lower workload, i.e. fewer POST messages.

A fourth possibility is that the cars take different routes altogether. Using this analogy, we explore each cause below.

Case 1: Fewer cars — a drop in the number of requests

Our telemetry can only show that queueing delay fell; it cannot show why. To test whether fewer requests are part of the explanation, we examined two independent, publicly available datasets: OpenRouter's usage rankings [52] and Cloudflare Radar crawler and web-traffic data as compiled by TechnologyChecker.io [53].

OpenRouter usage data. OpenRouter is an API marketplace that routes developer requests to hundreds of models; its rankings count prompt and completion tokens processed through its API, aggregated in daily UTC buckets [52]. Data run through 22 September 2026.

- No Anthropic model appears among the ten models with the most tokens processed over the trailing 30 days. The list is led by GLM 5.3 Flash (Z.ai, 52T tokens), GPT-5.6 Luna (OpenAI, 51.4T, +209% versus the prior 30 days), Hy4 preview (Tencent, 51T) and DeepSeek V4 Flash 0731 (48.1T, +46%); seven of the ten come from Chinese developers (Table 2) [52]. OpenRouter notes that GPT-5.6 Luna is the only closed model in the 30-day top ten by token volume [50].
- By share of text requests in the week beginning 14 September 2026, Anthropic ranked seventh with 2.7%, and its request count rose 3% week on week. DeepSeek led with 25.4%, followed by Google (18.6%) and OpenAI (17.0%, with requests down 30% week on week) (Figure 4) [52].
- Anthropic's agentic coding tool remains heavily used: Claude Code was the second-largest app by tokens in OpenRouter's most recent daily app ranking (954B tokens), behind Hermes Agent (1.61T) [52].
- Measured by spend rather than volume, OpenRouter's weekly wallet-share chart showed OpenAI models overtaking Anthropic models in the week of 7 September 2026, for the first time since February 2024; the same report notes that US models' combined share of token volume on OpenRouter fell from about 70% to about 30% between June 2025 and June 2026 [58]. Commentators have debated what OpenRouter's steeply rising token totals imply for the AI investment cycle [33], and prediction-market forecasts for Anthropic's OpenRouter share have fallen as cheaper rivals gained ground [32].

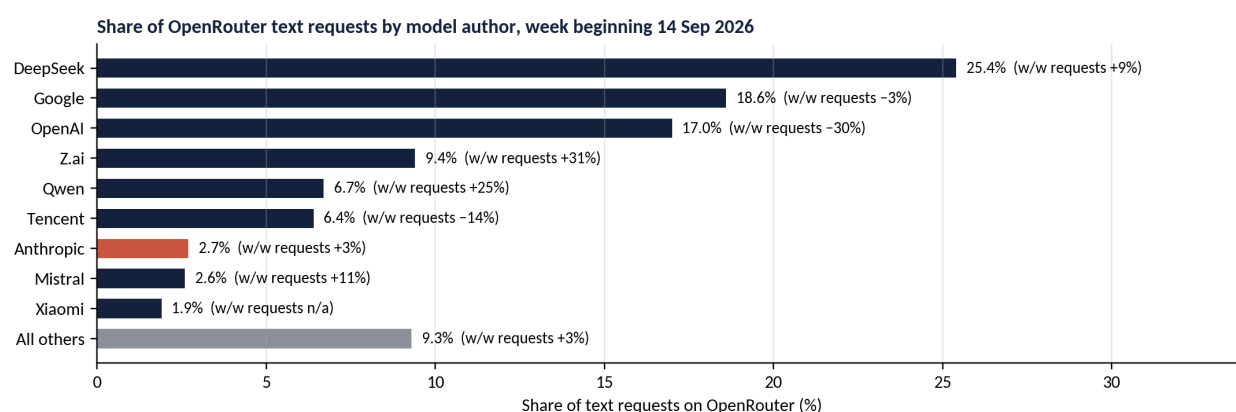
Table 2. Top 10 models on OpenRouter by tokens processed, trailing 30 days to 22 September 2026

Rank	Model	Developer	Tokens (30 days)	Change vs prior 30 days
1	GLM 5.3 Flash	Z.ai (China)	52.0T	new
2	GPT-5.6 Luna	OpenAI (US)	51.4T	+209%
3	Hy4 preview	Tencent (China)	51.0T	new
4	DeepSeek V4 Flash 0731	DeepSeek (China)	48.1T	+46%
5	MiMo-V2.5	Xiaomi (China)	28.0T	-1%

Rank	Model	Developer	Tokens (30 days)	Change vs prior 30 days
6	DeepSeek V4.1 Flash	DeepSeek (China)	25.9T	new
7	Hy3	Tencent (China)	20.2T	-37%
8	DeepSeek V4 Flash 0423	DeepSeek (China)	19.6T	-23%
9	Nemotron 3 Ultra (free)	NVIDIA (US)	18.5T	+47%
10	Ox Alpha	Stealth (undisclosed)	15.7T	+36%

Source: OpenRouter LLM Rankings, "This Month" view [52]. Rankings data © OpenRouter, Inc., licensed under CC BY 4.0.

Figure 4. Share of OpenRouter text requests by model author, week beginning 14 September 2026



Source: OpenRouter LLM Rankings, Market Share view, usage data through 22 September 2026 [52]. Rankings data © OpenRouter, Inc., licensed under CC BY 4.0. Percentage changes are week-on-week changes in each author's request count.

Caveats. OpenRouter states that its rankings measure adoption, not quality; that they describe only traffic routed through OpenRouter, not the whole market or usage of a provider's own API; and that token volume is not a count of requests, users or spend [52]. OpenRouter users are among the most price-sensitive in the market, and most Anthropic traffic flows through channels OpenRouter cannot see: Anthropic's first-party API, Claude subscriptions, AWS Bedrock and Google Cloud Vertex AI.

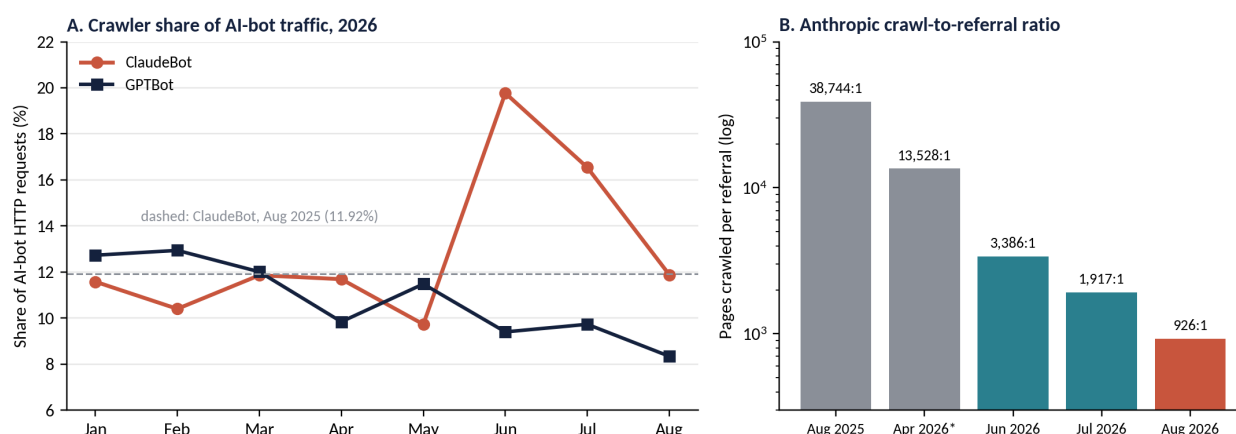
Crawler and web-traffic data. TechnologyChecker.io compiles Cloudflare Radar data on Anthropic's web crawlers and on Claude's web popularity; its latest update, published 3 September 2026, covers data through 31 August 2026 [53].

- ClaudeBot, Anthropic's training crawler, fell from a June 2026 peak of 19.77% of AI-bot HTTP traffic to 16.54% in July (revised) and 11.88% in August, within 0.04 points of its August 2025 level of 11.92%. GPTBot fell to 8.34%, so ClaudeBot still leads it by about 1.4× (Figure 5A) [53].
- Anthropic's share of all bot traffic fell from 10.97% in July to 8.47% in August, while its search crawler, Claude-SearchBot, rose to 3.69% [53].
- Anthropic's crawl-to-referral ratio halved from 1,917:1 in July to 926:1 in August, compared with 38,744:1 in August 2025, and claude.ai entered Cloudflare's top-ten referrer table for the first time, at 0.30% of crawler-observed referrals (Figure 5B) [53].
- Claude has ranked #2 in Cloudflare Radar's Generative AI service ranking, which is based on 1.1.1.1 DNS traffic, on every day from 22 January to 31 August 2026 (222 days), behind only ChatGPT [53].

Table 3. Anthropic crawler and web-traffic indicators (Cloudflare Radar)

Metric	Aug 2025	Jul 2026	Aug 2026
ClaudeBot share of AI-bot traffic	11.92%	16.54%	11.88%
GPTBot share of AI-bot traffic	12.25%	9.73%	8.34%
Claude-SearchBot share	—	3.54%	3.69%
Anthropic share of all bot traffic	below 1.76%	10.97%	8.47%
Anthropic crawl-to-referral ratio (lower = more referrals)	38,744:1	1,917:1	926:1
claude.ai share of crawler-observed referrals	not in top 10	not in top 10	0.30%
Claude rank, Cloudflare Radar Generative AI category	—	#2	#2

Source: Cloudflare Radar, as compiled by TechnologyChecker.io, data pulled 3 September 2026 [53]. July 2026 ClaudeBot share shown as revised by Cloudflare (originally reported as 16.28%).

Figure 5. Anthropic crawler share and crawl-to-referral ratio

Source: Cloudflare Radar, as compiled by TechnologyChecker.io [53]. Panel A: share of AI-bot HTTP requests by user agent, calendar months of 2026; July uses the revised 16.54% ClaudeBot figure. Panel B: pages crawled per referral sent (log scale); the April 2026 bar is the 13–20 April measurement window, other bars are calendar months.

Caveats. Crawler share measures how much Anthropic crawls the web to collect training and search data; it is not a measure of end-user demand for Claude. The compiler notes that monthly crawler shares swing widely and that the twelve-month change is close to zero [53]. The DNS-based ranking is a rank, not a volume, and TechnologyChecker.io states that it has no first-party Anthropic detection of its own [53].

If usage has genuinely reduced, token consumption has reduced with it. Because the revenue model is based on price per token, revenue would then be growing at a slower rate or declining.

Case 2: Bigger highways — capacity expansion

This is a real possibility, and it is what the extreme capex and data-center buildout signals. The real question is how much capacity expansion is actually needed to ensure that incoming traffic is handled well. Overbuilding while Case 1 is also true would lead to large capex increases with revenue not catching up. In our analogy, this means empty lanes on the highway. The capex spent on building these empty lanes needs to be justified in some way. Is it justifiable?

Case 3: Better traffic management — server optimisation

Best practice here involves autoscaling and proper utilisation of all available resources. This is an area of focus for many companies, since it is likely to deliver the best return on investment.

Case 4: Different routes — SLM adoption and distributed computing

The cars take many different routes, and instead of one big highway there are many roads. This is the case for small-language-model adoption and distributed computing described in Sections 2 and 4.

Essentially, if Cases 1 and 2 are true at the same time, this will likely lead to unnecessary capex investment. If Cases 1, 2 and 4 are true at the same time, the impact will be harsher.

D. Conclusion

In our view, the generative AI infrastructure boom is entering a more cost-conscious phase. Meta's move toward acting as a compute landlord, exploring a lease of up to \$10 billion of capacity to Anthropic [3], may mark a turning point in the centralized infrastructure buildout. Zuckerberg's rejection of an industry-wide slowdown [13] could, in our interpretation, also work to Meta's competitive advantage: Meta funds its buildout from advertising cash flow, while venture-funded frontier labs carry large compute commitments. Separately, the chairman of the US Federal Trade Commission, speaking in a personal capacity and without naming Anthropic, said that everyone should be "deeply suspicious" of AI companies that seek antitrust exemptions while lobbying for new regulation, days after Anthropic's CEO called for antitrust permission for rivals to coordinate a slower pace of development [54].

Taken together, the Stanford research [18][20], Gartner's forecast [22], the ServeGen production data [23], the OpenRouter usage data [52] and our single-vantage-point telemetry suggest that enterprise AI is partially decentralizing toward smaller, cheaper and more controllable models, especially for simple, high-volume work. If that trend continues and extends to agentic workloads, a meaningful share of centralized data-center capacity could become underused. Indicators that would test this view include hyperscaler capex revisions, compute lease pricing, the outcome of the Meta-Anthropic talks, and Anthropic's Q3 2026 results.

Notes on Methodology

- **Data provenance.** The latency, time-to-first-token and error-rate data behind Figures 1 to 3 were measured, not modelled. They were collected from a single user's vantage point: one account, one region and one network location. The word "model" in this report refers only to the M/D/1 queueing model used to derive the congestion index from those measurements; it does not describe the nature of the data. Measurements taken from another user's vantage point may differ materially, particularly absolute latency and HTTP 429 error rates, which depend on each account's quotas and rate-limit tier. Month-to-month trends are more likely to be reproducible across observers than absolute levels.
- **Definition of the index.** The index is estimated server utilisation (ρ) divided by its average over January–February 2026, multiplied by 100. It is a relative measure, not utilisation itself. The underlying aggregate estimated utilisation falls from about 0.83 in January 2026 to about 0.61 in September 2026 (Figure 1).
- **Formula and source.** Utilisation is estimated as $\rho = 2q / (1 + 2q)$, where $q = W / D - 1$, W is observed latency and D is the base service time. This is rearranged from the M/D/1 queueing model of average time to first token in DistServe (Zhong et al., OSDI 2024; arXiv:2401.09670) [51], in which average TTFT = $D + RD^2 / (2(1 - RD))$ and $\rho = RD$. The model assumes random (Poisson) request arrivals, a constant processing time per request, and a single first-come, first-served queue. The original authors note that variation in request length causes deviations from the model [51].
- **Latency measure used.** The formula is applied to mean full response latency (Figure 2). The published model describes time to first token, so this is an approximation. Recalculating the index from the recorded time-to-first-token values, with network round-trip time subtracted, would align it more closely with the published model.
- **Base service time (D).** $D = 546$ ms, fitted to the measured latency series in Figure 2 and scaled by each platform's January latency (557 ms for AWS Bedrock, 538 ms for Google Cloud Vertex AI and 543 ms for the Anthropic API). D is held constant for the whole period. If model or hardware changes lowered the true base service time during the period, a fixed D overstates the decline in congestion. In future iterations, D should be estimated for each period (for example, as a low percentile of time to first token during the quietest hours) and reset whenever the served model version changes.
- **Smoothing.** The formula is applied to 7-day trailing average latency rather than daily values. The formula becomes highly sensitive when latency approaches D , so applying it to daily values would mainly amplify day-to-day noise.
- **Platform-level aggregation.** Values are aggregated per platform across models and do not isolate the performance of any single model.
- **Short-lived events are smoothed out.** Figure 2 shows daily values together with 7-day averages, and Figure 3 uses 7-day averages only. At this resolution, most service incidents and outages, which typically last minutes to a few hours, are largely absorbed: a 25-minute outage affects less than 2% of a day's requests, and a 7-day average dilutes it further. This is a property of the aggregation, not of the underlying measurements. Only longer or more severe incidents are likely to remain visible as bumps in daily values. Lasting changes, such as a shift in latency after a model version change, are not removed by smoothing; they appear as a gradual step spread over roughly a week. The figures are not aligned to specific published incidents or launch dates, and individual events should be verified against the raw request logs rather than the averaged series.

Limitations of the M/D/1 methodology

The M/D/1 model is a deliberately simple approximation, and the congestion index inherits its limitations. We acknowledge the following:

1. **Arrival process.** The model assumes random (Poisson) arrivals. Production LLM traffic is burstier: ServeGen found coefficients of variation above 1 for language-model request arrivals [23]. For a given utilisation, bursty

arrivals produce longer queues than M/D/1 predicts, so inverting the model tends to overstate ρ . The level of the index is therefore biased; its trend is less affected as long as burstiness is broadly stable over time.

2. **Service time.** The model assumes every request takes the same time to process. Real prompt and output lengths vary widely, and DistServe's own authors note that non-uniform request lengths cause deviations from the M/D/1 results [51]. Variable service times also lengthen queues at a given utilisation, again biasing ρ upward.
3. **Server structure.** M/D/1 describes one server processing one request at a time, first come, first served; DistServe applies it to a disaggregated prefill instance [51]. Commercial endpoints use continuous batching, many replicas behind load balancers, autoscaling and priority tiers. Queueing delay in such pooled systems relates to utilisation very differently, so ρ here is an indicator, not the provider's actual server utilisation.
4. **What is measured.** The model describes time to first token, but the index is computed from mean full response latency, which also includes decoding time, network round-trip time and client-side overhead. Changes in the mix of output lengths, or in network paths, can move the index without any change in server load.
5. **Identification.** A lower index is consistent with fewer requests (Case 1), added capacity (Case 2), faster serving software or hardware (Case 3), traffic routed elsewhere (Case 4), or any combination of them. The index alone cannot distinguish between these causes, which is why Case 1 is tested against external data.
6. **Admission control and product changes.** Rate limits (HTTP 429 responses) shed load before it queues, and providers change quotas, caching and model versions over time; for example, Anthropic raised API rate limits for its Opus models in May 2026 [10]. Such changes alter observed latency independently of underlying utilisation.
7. **Sensitivity to the base service time.** The formula becomes unstable as W approaches D , and results depend heavily on the assumed value of D (see Base service time above).
8. **Single vantage point.** All inputs come from one user's vantage point (see Data provenance), so the index describes that observer's experience, not the platform as a whole.

The congestion index should therefore be read as a directional, relative indicator of the queueing delay experienced by one observer, not as a measurement of provider utilisation, capacity or demand.

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